

CWRC, IDV, and Time Dilation

Wave Language, Internal Volume, and the Causal Chain

Extending the δ -chain with IDV-driven metric coupling

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March 2026

Abstract

This document extends the CWRC Bridge with two results that emerged from cross-session analysis. First, the Internal Dimensional Volume (IDV) is connected to the time dilation result derived in CRF Entropy Dynamics: the causal chain $\text{IDV} \uparrow \Rightarrow \text{local time} \uparrow \Rightarrow \text{metric curvature} \uparrow$ is a direct consequence of $H(t) = H_0/(1 + \alpha H_0 t)$. Second, the IDV-metric coupling mechanism is specified precisely: the outer product is taken on the 3D subspace only, consistent with the $\text{SO}(3)$ derivation of three spatial dimensions (Session 5). V17 simulator design follows from these two results.

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1 The Causal Chain: IDV to Time Dilation

The CRF Entropy Dynamics document derived:

$$\frac{dH}{dt} = -\alpha H^2 \implies H(t) = \frac{H_0}{1 + \alpha H_0 t} \implies \frac{d\tau_M}{dt} \propto H(t) \rightarrow 0$$

This was derived for a Mind M approaching its entropy boundary. The same structure applies to any node with accumulated R-event history.

IDV–Time Dilation Chain

In the simulator, for node i :

$$\delta \rightarrow R\text{-events} \rightarrow N_{\text{eff},i} \uparrow \rightarrow \text{mass}_i = 1 + \kappa N_{\text{eff},i} \uparrow \rightarrow \Omega_i = \text{mass}_i - 1 \uparrow$$

$$\Omega_i \uparrow \equiv D_{\text{KL}}(C_i \| P_{\text{bg}}) \uparrow \Rightarrow H(C_i) \downarrow \Rightarrow \frac{d\tau_i}{dt} \downarrow$$

$$\frac{d\tau_i}{dt} \downarrow \Rightarrow \text{node } i \text{ resolves less per external sweep} \Rightarrow \text{metric curvature at } i \uparrow$$

IDV encodes the same information:

$$\text{IDV}_i = H(P_i) \times (1 + \phi\text{-diversity}) \times (1 + |\nabla\Psi|_i) \times \ln(1 + \text{mass}_i)$$

As mass $\uparrow$: $\ln(1+\text{mass})$ grows and $H(P_i)$ falls (Born rule converges P_i). The product captures the node's *internal richness* — high when many patterns are accessible and history is deep.

$\text{IDV}_i \uparrow \Rightarrow$ node has more internal states $\Rightarrow$ local time flows slower $\Rightarrow$ marble with more “space inside.”

Status: **Derived** (chain follows from HDynamics + EIC Bridge).

2 Why the Metric Coupling Uses 3D, Not 8D

IDV-Metric Coupling Specification

The update to the metric tensor at node i after an R-event:

$$G_{\text{new}}^{(i)} += \lambda \cdot \text{IDV}_i \cdot \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \cos(\Phi_j - \Phi_i) \cdot \mathbf{r}_{ij}^{(3)} \otimes \mathbf{r}_{ij}^{(3)}$$

where $\mathbf{r}_{ij}^{(3)} = (X_j - X_i)_{[1:3]}$ is the 3-dimensional component only.

The remaining block $G_{[4:8,4:8]}^{(i)}$ is updated by the existing SINDy pruning (not by IDV).

Why the 3D slice, not full 8D?

Session 5 derived that $n = 3$ is the minimum dimension in which $\text{SO}(3)$ has stable irreducible representations under δ -operation. This means: the geometry that *persists* is 3D.

When the IDV outer product strengthens $G_{[1:3,1:3]}$, the random walker (used to estimate d_s) becomes preferentially trapped in the 3D subspace. High-IDV nodes pull d_s toward 3 first; the effect propagates to neighbours through the metric coupling — the Marble Effect.

Marble Effect

A high-IDV node (heavy, old, many R-events) has a 3D-dominant metric. Its neighbours couple to it via $\cos(\Phi_j - \Phi_i)$ (phase alignment). Aligned neighbours inherit the 3D metric structure. This creates a propagating wave of $d_s \rightarrow 3$ — not imposed, but grown from the nodes with the deepest history.

3 Product Form: Path to Formal Derivation

CRF AI identified two steps needed to raise IDV from Hypothesis to Lemma.

3.1 Step α : Density as $N_{\text{distinguishable}}$

Open: IDV as Density of Possibility

Claim: $\text{IDV}_i \propto N_{\text{distinguishable},i}$ where $N_{\text{distinguishable}}$ is the number of states node i can actually distinguish given its current C_i .

Derivation path:

$$N_{\text{distinguishable}} = e^{H(P_i)} \quad (\text{from information theory: } H = \ln N)$$

Phase diversity multiplies this by the number of *phase-distinct* configurations accessible to neighbours. $|\nabla\Psi_i|$ multiplies by the number of speciation-distinct neighbours. $\ln(1 + \text{mass}_i)$ multiplies by the log of resolved events.

If these factors are *information-independent* (each measures a different dimension of P -space), their product is the correct count of distinguishable states.

Status: **Open** — requires proof of independence.

3.2 Step β : Multiplicative Form from Independence

Open: Information Independence of IDV Factors

For the product form to be valid, we need:

$$I(H(P_i); \phi\text{-diversity}) \approx 0 \quad \text{and} \quad I(\phi\text{-diversity}; |\nabla\Psi_i|) \approx 0$$

Intuition: H measures *within-node* entropy; phase diversity measures *between-node wave structure*; $|\nabla\Psi|$ measures *speciation boundary sharpness*; $\ln(1 + \text{mass})$ measures *temporal depth*. These are orthogonal perspectives on the same node.

Analogous to Lemmas 13–15 in Session 9 (independent derivations of complementary quantities). Formal proof requires showing that the mutual information between factors is bounded by ε_0 .

Status: **Open** — but strongly supported by the structure of the δ -chain.

4 V17 Simulator Design

V17 extends V16 (canonical base: $R^2 = 0.906$, $d_s = 3.008$) with:

V17 Components

1. Φ field (wave layer):

$$\Phi_i \leftarrow \Phi_i + \eta \sum_{j \in \mathcal{N}(i)} D_{\text{KL}}(P_i \| P_j) \cdot \sin(\Phi_j - \Phi_i)$$

2. Ψ_i modes (pattern identity):

$$\Psi_i \leftarrow 0.9 \Psi_i + 0.1 \bar{\Psi}_{\mathcal{N}(i)} + \varepsilon_0 \xi_i$$

Selective coupling: $\kappa_{ij} = D_{\text{KL}}(P_i \| P_j) \cdot e^{-|\Psi_i - \Psi_j|/\sigma}$

3. IDV computation per node per sweep.

4. IDV-driven metric update (3D slice):

$$G_{[1:3,1:3]}^{(i)} += \lambda \cdot \text{IDV}_i \cdot \cos(\Phi_j - \Phi_i) \cdot \mathbf{r}_{ij}^{(3)} \otimes \mathbf{r}_{ij}^{(3)}$$

5. Three observables beyond V16:

- $\overline{V^{\text{int}}}(t)$ — Internal Dimensional Inflation
- $\overline{d^{\text{int}}}(t)$ — local effective dimension per node
- Speciation histogram of Ψ_i — matter type distribution

Predicted results from V17: If IDV-metric coupling works as designed, d_s (global) will be pulled toward 3 faster than V16 alone, starting from nodes with highest IDV (cores) and propagating outward. $\overline{V^{\text{int}}}$ will increase monotonically while $d_s \approx 3$ — internal expansion with geometric stability. This would be the first direct simulation of the marble intuition.

5 Complete Status Table

Result	Status	Note
CWRC = re-expression of δ -chain	Derived	$J = G, \Phi = D_{\text{KL}}, \Psi = P_M$
Kuramoto from CRF Syn-chronicity	Derived	Special case
IDV causal chain via HDynamics	Derived	$\Omega \uparrow \Rightarrow H \downarrow \Rightarrow d\tau \downarrow$
IDV-metric coupling (3D slice)	Derived	SO(3) forces 3D
IDV product form	Hypothesis	Independence not proved
IDV as $N_{\text{distinguishable}}$	Open	Step α
Multiplicative form proof	Open	Step β
Marble Effect in V17	Open	Simulation pending
IDI observable	Open	V17 will measure

The marble does not expand outward. It becomes capable of more distinctions within the same boundary. That is what history does to a node.